

# Patrick O'Brien

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## PROFESSIONAL SUMMARY

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Software Engineer with a B.S. in Computer Science (2025) and hands-on experience building full-stack web applications and production automation tools. Comfortable across the stack: React front ends, FastAPI/Flask APIs, SQL data layers, and Python automation against cloud APIs, with CI/CD pipelines and automated test coverage throughout.

## SKILLS

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**Languages:** Python, TypeScript, JavaScript, PowerShell, SQL, Java, C++, C, HTML/CSS

**Frameworks:** React, FastAPI, Flask, SQLAlchemy

**Tools & Technologies:** Git/GitHub, GitHub Actions (CI/CD), REST APIs, JWT Auth, pytest, Microsoft Graph API, Microsoft 365, Microsoft Entra ID, Azure

**Databases:** PostgreSQL, MySQL, SQLite, Microsoft Access

**Core Concepts:** Object-Oriented Programming, Data Structures, Web Development, Automated Testing

**Operating Systems:** Linux, Windows, macOS

## EXPERIENCE

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### IT Support Specialist, Software Automation Focus

Jun 2025 – Present

*Habitat America*

Annapolis, MD

- Building a Python automation tool using the Microsoft Graph API to fully automate user provisioning and deprovisioning across 100+ properties and 400+ users, eliminating manual account setup and reducing onboarding to a single automated workflow.
- Administer and maintain Microsoft 365, Microsoft Entra ID, and Exchange environments supporting 400+ users across 100+ managed properties.
- Diagnose and resolve technical issues across systems and applications, applying structured, engineering-driven problem-solving under time constraints to keep operations running smoothly.

### Database Administrator

Mar 2019 – May 2025

*Mike's Mowing (family business)*

Ridgely, MD

- Independently designed, built, and maintained a Microsoft Access database for a family-owned landscaping business, covering customer records, service history, and scheduling.
- Owned data integrity across the system: validating inputs, resolving inconsistencies, and structuring tables, queries, and forms to keep records accurate and reliably accessible over a 6-year period, while working full-time alongside coursework.

### Job Coach

Mar 2013 – Mar 2019

*Benedictine*

Ridgely, MD

- Provided on-the-job coaching and support for individuals with disabilities in workplace settings.

## PROJECTS

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### DartMetrics [github.com/pobriendev/dartmetrics](https://github.com/pobriendev/dartmetrics)

*FastAPI, PostgreSQL, SQLAlchemy, React, TypeScript, JWT, Playwright*

- Built a full-stack darts scoring platform (501, Cricket, Halve It) for play against friends or bots, centered on a pure, framework-free scoring engine that is fully unit-tested in isolation from the API and database.
- Stored every dart thrown as an individual event, deriving averages, checkout percentage, and 180 counts from raw events rather than stored aggregates, with database check constraints that reject impossible game states.
- Wrote 255 backend tests, 47 frontend tests, and Playwright end-to-end tests, run by a GitHub Actions CI pipeline on every push; deployed to production with a live demo at [dartmetrics.onrender.com](https://dartmetrics.onrender.com).

### Ticket Management System [github.com/pobriendev/ticket-system](https://github.com/pobriendev/ticket-system)

*React (Vite), FastAPI, PostgreSQL, JWT, GitHub Actions*

- Architected a full-stack helpdesk ticketing application with JWT authentication, role-based access control, and an automatic audit trail on every status and assignment change.
- Designed the data model and REST API (FastAPI + SQLAlchemy/Alembic) covering ticket CRUD, filtering, pagination, comments, and category management.
- Wrote 171 backend tests (~96% coverage) and 128 frontend tests with pytest and Vitest, and built a CI/CD pipeline (GitHub Actions) running tests and lint on every push, deploying on merge to main.

### Employee Provisioning Tool [github.com/pobriendev/employee-provisioning-tool](https://github.com/pobriendev/employee-provisioning-tool)

*Python, Microsoft Graph API, YAML*

- Built a Python CLI that automates Entra ID employee onboarding and offboarding end-to-end: account creation or role-account reuse, license assignment, group membership, and credential delivery, replacing a manual multi-step portal process.
- Designed app-only (client-credentials) authentication scoped to least-privilege Graph permissions, with dry-run mode, lockout-first offboarding, and a full audit log for every write action.

**Entra Stale Accounts** [github.com/pobriendev/entra-stale-accounts](https://github.com/pobriendev/entra-stale-accounts)

*Python, Microsoft Graph API, PyPI*

- Published a read-only CLI to PyPI (pip install entra-stale-accounts) that flags inactive Microsoft Entra ID accounts past a configurable threshold, including never-signed-in accounts.
- Generalized the tool from production account-lifecycle automation into a standalone package with a mocked test suite, CI via GitHub Actions, and automatic Graph API retry/backoff.

**EDUCATION**

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**Delaware State University**

*B.S. in Computer Science. GPA: 3.57.*

2025

**Delaware Technical Community College**

*A.S. in Computing and Information Science.*

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